I Can find even products

Level 3
Operations
Teacher 1

CCSS 3.OA.D.9, 3.OA.C.7 · **EE** M.EE.3.OA.8 · **CRA:** Concrete → Pictorial · **Flow:** Standard (3 new routine steps)

Lesson guidance:

Tip: Cubes: two equal parts. Build 4 rows of 6: 12 and 12, two equal parts, so it is even.

Tip: Row OR column. Say "row or column" every time. One even number is enough.

Watch: Row only. The mistake to stop today: an even column left white.

Lesson flow:

#	Section	Pupil page	Min
1	Prerequisite Check	Prerequisite Check	6
2	Number sense	Number sense	5
3	Spaced retrieval	Mini whiteboards	4
4	Chaining bridge	Mini whiteboards	2
5	I Do: the anchor chart	Anchor chart	10
6	We Do: Guided Practice	Lesson sheet	7
7	Check Question	Mini whiteboards	3
8	You Do: Independent Practice	Practice 1	9
9	Word problem	Word problem	5
10	Plenary and exit slip	Exit slip	5
	Transitions	between pages	4
	Total		**60**

Materials:

- ☐ Interlocking cubes: a teacher set of 60
- ☐ Anchor chart (a 2 × 4 window), A3 or projected
- ☐ Exit slips, cut in half
- ☐ Mini whiteboards and pens
- ☐ Pupil packet, one per pupil

Vocabulary:



even

makes pairs, none left over



odd

one is left over

×	3
4	12

product

the answer when you multiply

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I Can find even products

Level 3
Operations
Teacher 2

What students need to know:

#	Skill or Concept	Atom Type	Teacher Notes
1	**Even numbers:** the count by 2s (2, 4, 6, 8 …)	Category · Pre-requisite	Prerequisite Check *a*.
2	**Facts of 2 and 4:** 4 × n is 2 × n twice	Fact · Pre-requisite	Prerequisite Check b–e.
3	**Read the chart:** row meets column	Fact · Pre-requisite	Prerequisite Check f–g.
4	**Even row:** every product is even	Comparative	Anchor chart, the even rows.
5	**Why:** an even row splits into two equal parts	Conceptual	Cubes: 4 rows of 6, 12 and 12.
6	**One even factor is enough: row OR column**	**Category**	**Critical Atom. Routine step 2.**
7	Odd × odd is odd (contrast)	Comparative	Anchor chart step 3; every window.

What students do:

Step	Task / Routine atom	Tag	Check it with
1	Shade the even rows.	New	Guided 1, exit *a*
2	**Shade the even columns too.**	**New**	**Guided 2, Check Question, exit b**
3	Both odd? Leave it white.	New	Practice last row, exit c

Rows tagged New: 3 → Flow: Standard. One chunk: one even row or column number makes the product even.

How to teach:

Show, then name. Build the even row of the chart with cubes and split it into two equal parts, then shade the even rows on the window, then the even columns. Odd × odd is shown last, as a contrast, never as a second rule.

Board: Window rows 4, 5 × columns 6, 7, 8, 9.

Common mistakes:

Mistake	Step	Looks like	Fix	Caught by
Row only	2	20, 24, 28	Row OR column? Check both.	Check Question A
Odd × odd = even	3	18, 20, 24, 28, 15, 21	Count the pairs: one is left over.	Check Question B
Tens digit decides	1	30 left white (its tens digit is odd)	Cover the tens. Read the ones.	Practice 1, a white box
Any odd factor is odd	2	5 × 6 left white	One even number is enough.	Exit b
Columns only	1	The 6 column shaded, the 6 row white	Rows first, then columns: finger on each header.	Exit *a*

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I Can find even products

Level 3
Operations
Reference

Reference key:

Pages and order

Page	Tab	Score
P0	Prerequisite	one per question
P1	Number sense	/10
P2	Anchor chart	none (a poster)
P3	Lesson 1	Independent cells
P4	Practice n	every cell
P5	Word problem	/1
P6	Exit slip	/3 a half
T1–T5	Teacher n	—
T6	Answer Key	—

Fade levels

Label	Means
★	Guided: hints on, step 1 traced
★★	Independent: hints off
★★★	Last row: the edge cases

Data loop

● **Most:**	Go on.
◑ **About half:**	Short reteach, then go on.
○ **Fewer than half:**	Stop. Reteach with a concrete example.

Abbreviations

Short	In full
MWBs	mini whiteboards
ELL	English language learners
SPED	special education students
RCDWS	Read, Circle, Draw, Write, Solve
CRA	Concrete-Representational-Abstract

Routing ids

No site lesson exists yet for these prerequisites: T6 names a CCSS code and a short title (V22 Appendix H).

Answer slots

Shape	Used for
_____	a number after =
☐	a part inside an expression
☐	a tick (routine panel, exit slip)

Icons

Icon	Means
	Tip
	Feedback
	Watch
	Research
	Critical Atom
	Language tip
	Why it works
	Board
1 2 3	The pupil Steps (accent)

Band labels

Pupil: Vocabulary · Remember · Guided Practice · Steps · Independent Practice · Mixed Review · Number sense: A / B · Workspace · Another example · Say · Rule.

Teacher: Lesson guidance · Lesson flow · Materials · What students need to know · What students do · How to teach · Common mistakes · Section n · Prerequisite routing · Accepted answers · Answers at a glance.

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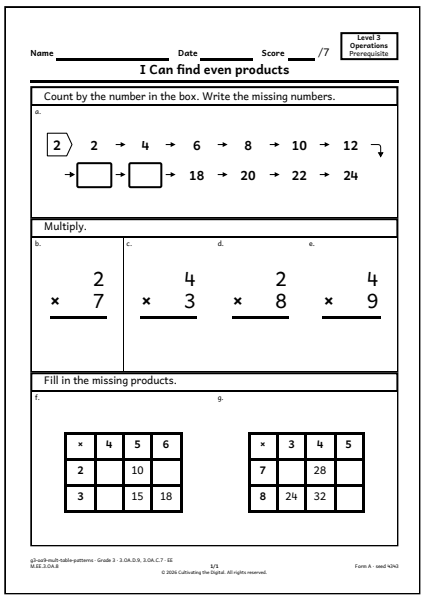
I Can find even products

Level 3 Operations Teacher 3

Section 1 · Prerequisite Check (6 min)

> **Tip: Route before teaching.** Mark it while pupils do the Number sense page.

1. **Do:** Hand out the Prerequisite Check, face down.
2. **Say:** "Do these on your own. Try every one."
3. **Do:** Walk and mark as pupils finish.
   Note: Expected: **a. 14, 16 · b. 14 · c. 12 · d. 16 · e. 36 · f. 8, 12, 12 · g. 21, 35, 40**. Routing is on T6.



Name _____ Date _____ Score ___ /7 — Level 3 Operations Prerequisite

I Can find even products

Count by the number in the box. Write the missing numbers.

a.

2 → 2 → 4 → 6 → 8 → 10 → 12 → ☐ → ☐ → 18 → 20 → 22 → 24

Multiply.

b. 2 × 7 = ____

c. 4 × 3 = ____

d. 2 × 8 = ____

e. 4 × 9 = ____

Fill in the missing products.

f.

×	4	5	6
2		10	
3		15	18

g.

×	3	4	5
7		28	
8	24	32	

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Section 2 · Number sense (5 min)

1. **Do:** Hand out the Number sense page. Band A, then band B.
2. **Ask:** "What is 10 more than 47?"
   Note: Expected: **57**.
3. **Ask:** "What is 10 more than 877?"
   Note: Expected: **887**.

Section 3 · Spaced retrieval (4 min)

1. **Do:** Pupils hold mini whiteboards. Write each fact on the board.
2. **Ask:** "5 × 5 = ?"
   Note: Expected: **25**.
3. **Ask:** "2 × 10 = ?"
   Note: Expected: **20**.
4. **Ask:** "7 × 8 = ?"
   Note: Expected: **56**.

Section 4 · Chaining bridge (2 min)

1. **Say:** "Even numbers make pairs."
2. **Say:** "Today we find them in the times table."

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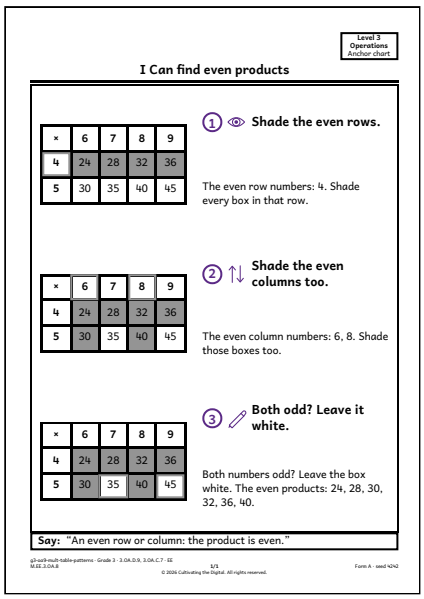
I Can find even products

Level 3 Operations Teacher 4

Section 5 · I Do: the anchor chart (10 min)

Watch: Row only. Point to the row number AND the column number, every box.

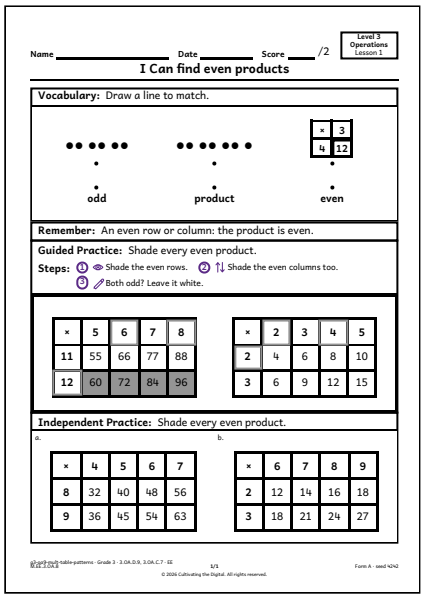
1	**Do:**	Put the anchor chart up: rows 4, 5, columns 6, 7, 8, 9.
2	**Do:**	Build 4 rows of 6 cubes.
3	**Say:**	"4 is even. Every box in that row is even."
	Note:	Step 1: shade the row.
4	**Ask:**	"What about the 6 column?"
	Note:	Expected: even. Shade the column.
5	**Say:**	"5 is odd. 6 is even. So 30 is even."
	Note:	Step 2, the Critical Atom: one is enough.
6	**Say:**	"5 and 9 are both odd. So 45 is odd."
	Note:	Leave it white. Contrast only.
7	**Do:**	Point to 4 × 6 = 24: the cubes. Split them into two equal parts.
8	**Say:**	"24 is 12 and 12: two equal parts."
	Note:	Why it is even: it splits into two equal parts.



Section 6 · We Do: Guided Practice (7 min)

Feedback: Fix it at once. Fix: "Check the row AND the column."

1	**Say:**	"Finger on the row number. Finger on the column number."
2	**Ask:**	"Is one of them even?"
	Note:	Listen for "row or column".
3	**Do:**	Pupils shade the first window with you, row by row.
	Note:	Expected ★: **66, 88, 60, 72, 84, 96**.
4	**Say:**	"11 is odd. 6 is even. So 66 is even."
	Note:	The odd row: its even column.
5	**Do:**	Pupils shade the second window alone, row by row.
	Note:	Expected ★: **4, 6, 8, 10, 6, 12**.
6	**Ask:**	"Which boxes stay white?"
	Note:	Both numbers odd: **9, 15**.
7	**Do:**	Pupils do the two Independent windows under the Guided cells.
	Note:	Circulate: a white box with an even number?



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I Can find even products

Level 3
Operations
Teacher 5

Section 7 · Check Question (3 min)

1. **Do:** Draw the window rows 3, 4 × columns 5, 6, 7 three times. Shade A: 20, 24, 28 · B: 18, 20, 24, 28, 15, 21 · C: 18, 20, 24, 28.

2. **Ask:** "Which shading is right? Show fingers."
   Note: Correct: **C 18, 20, 24, 28**. A: Row only · B: Odd × odd = even

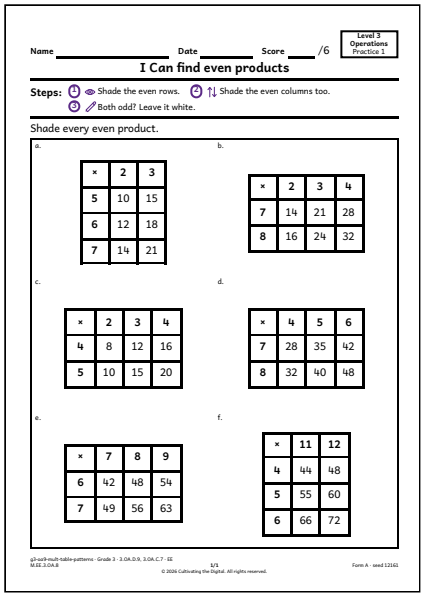
● **Most:** Go on to You Do.
◑ **About half:** Re-show the odd row with an even column, then one more.
○ **Fewer than half:** Back to the cubes: 4 rows of 6, two equal parts.

Section 8 · You Do: Independent Practice (9 min)

Watch: The last row. Walk to the last row first. Practice 1: 6, 7 × 7–9: 54 and 56 are even (tens digit odd), 63 stays white; 4–6 × 11, 12: 72 is even (tens digit odd), 55 stays white.

1. **Do:** Pupils do Practice 1. The Steps are at the top of the page.

2. **Do:** Circulate. Point at every white box.
   Note: Ask: "Row odd? Column odd?"



Name _____ Date _____ Score ___ /6 — Level 3 Operations Practice 1

I Can find even products

Steps: Shade the even rows. Shade the even columns too. Both odd? Leave it white.

Shade every even product.

a.

×	2	3
5	10	15
6	12	18
7	14	21

b.

×	2	3	4
7	14	21	28
8	16	24	32

c.

×	2	3	4
4	8	12	16
5	10	15	20

d.

×	4	5	6
7	28	35	42
8	32	40	48

e.

×	7	8	9
6	42	48	54
7	49	56	63

f.

×	11	12
4	44	48
5	55	60
6	66	72

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Section 9 · Word problem (5 min)

1. **Do:** Read the story aloud two times. Pupils tick each box.

2. **Ask:** "How many apples? Even or odd?"
   Note: Expected: **28**, **even**.

3. **Say:** "4 is even. So 28 is even."
   Note: The Say line on the page: one even number is enough.

Section 10 · Plenary and exit slip (5 min)

1. **Say:** "If one number is even, the product is even."
   Note: The Remember line. Say it twice.

2. **Do:** Hand out the exit slip. No help. Sort by the boxes ticked.
   Note: a. **30, 30, 36, 42** · b. **22, 44, 24, 36, 48** · c. **56, 64, 72, 72**.

3. **Say:** "Tomorrow we use rules to make number patterns."

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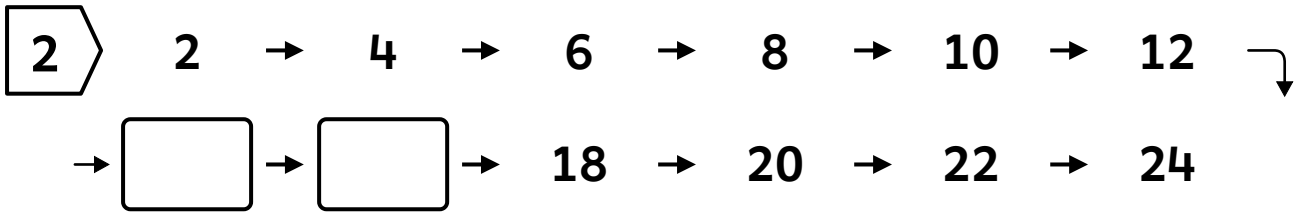
Name _____ Date _____ Score _____ /7

Level 3
Operations
Prerequisite

I Can find even products

Count by the number in the box. Write the missing numbers.

a.



Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline \end{array}$$

c.

$$\begin{array}{r} 4 \\ \times\ 3 \\ \hline \end{array}$$

d.

$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline \end{array}$$

e.

$$\begin{array}{r} 4 \\ \times\ 9 \\ \hline \end{array}$$

Fill in the missing products.

f.

×	4	5	6
2		10	
3		15	18

g.

×	3	4	5
7		28	
8	24	32	

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Name _____ Date _____ Score _____ /10

Level 3
Operations
Number sense

I Can find even products

Number sense: A

1 Write the missing number.

10 more than **47** is _____ .

2 Write the missing number.

10 less than **20** is _____ .

3 Write <, > or = in the circle.

72 ◯ **35**

4 Write how many of each place.

39 = ☐ tens ☐ ones

5 Write the missing number.

1 more than **33** is _____ .

Number sense: B

6 Write the missing number.

10 more than **877** is _____ .

7 Write the missing number.

10 less than **305** is _____ .

8 Write <, > or = in the circle.

544 ◯ **565**

9 Write how many of each place.

84 = ☐ tens ☐ ones

10 Write the missing number.

1 more than **103** is _____ .

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Level 3
Operations
Anchor chart

I Can find even products

×	6	7	8	9
4	24	28	32	36
5	30	35	40	45

1 Shade the even rows.

The even row numbers: 4. Shade every box in that row.

×	6	7	8	9
4	24	28	32	36
5	30	35	40	45

2 Shade the even columns too.

The even column numbers: 6, 8. Shade those boxes too.

×	6	7	8	9
4	24	28	32	36
5	30	35	40	45

3 Both odd? Leave it white.

Both numbers odd? Leave the box white. The even products: 24, 28, 30, 32, 36, 40.

Say: "An even row or column: the product is even."

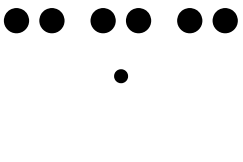
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Name _____ Date _____ Score _____ /2

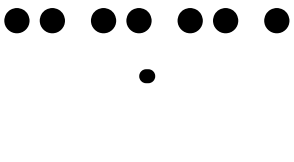
Level 3 Operations Lesson 1

I Can find even products

Vocabulary: Draw a line to match.



odd



product

×	3
4	12

even

Remember: An even row or column: the product is even.

Guided Practice: Shade every even product.

Steps: (1) Shade the even rows. (2) Shade the even columns too.
(3) Both odd? Leave it white.

×	5	6	7	8
11	55	66	77	88
12	60	72	84	96

×	2	3	4	5
2	4	6	8	10
3	6	9	12	15

Independent Practice: Shade every even product.

a.

×	4	5	6	7
8	32	40	48	56
9	36	45	54	63

b.

×	6	7	8	9
2	12	14	16	18
3	18	21	24	27

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Name _____ Date _____ Score _____ /6

Level 3 Operations Practice 1

I Can find even products

Steps: (1) Shade the even rows. (2) Shade the even columns too.
(3) Both odd? Leave it white.

Shade every even product.

a.

×	2	3
5	10	15
6	12	18
7	14	21

b.

×	2	3	4
7	14	21	28
8	16	24	32

c.

×	2	3	4
4	8	12	16
5	10	15	20

d.

×	4	5	6
7	28	35	42
8	32	40	48

e.

×	7	8	9
6	42	48	54
7	49	56	63

f.

×	11	12
4	44	48
5	55	60
6	66	72

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Name _____ Date _____ Score _____ /1

Level 3
Operations
Word problem

I Can find even products

Read the story. Draw it. Solve.

There are 4 bags.

Each bag has 7 apples.

How many apples? Is it even or odd?

There are _____ apples.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

☐ × ☐ = ☐

Say: "Is 4 or 7 even? The product is _____."

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Name _____ Score _____ /3

Level 3 Operations Exit slip

I Can find even products

Shade every even product.

a.

×	5	6	7
5	25	30	35
6	30	36	42

b.

×	2	3	4
11	22	33	44
12	24	36	48

c.

×	7	8	9
8	56	64	72
9	63	72	81

☐ I can do this. ☐ I need one more day.

Name _____ Score _____ /3

Level 3 Operations Exit slip

I Can find even products

Shade every even product.

a.

×	5	6	7
5	25	30	35
6	30	36	42

b.

×	2	3	4
11	22	33	44
12	24	36	48

c.

×	7	8	9
8	56	64	72
9	63	72	81

☐ I can do this. ☐ I need one more day.

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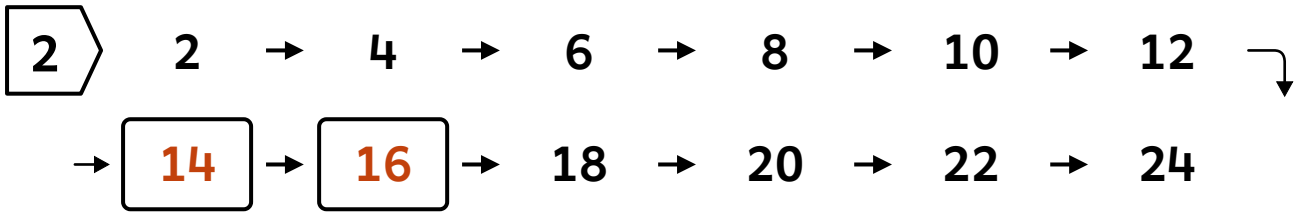
Answer Key _____ **Date** _____ **Score** _____ /7

Level 3 Operations Answer Key

I Can find even products

Count by the number in the box. Write the missing numbers.

a.



Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline 14 \end{array}$$

c.

$$\begin{array}{r} 4 \\ \times\ 3 \\ \hline 12 \end{array}$$

d.

$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline 16 \end{array}$$

e.

$$\begin{array}{r} 4 \\ \times\ 9 \\ \hline 36 \end{array}$$

Fill in the missing products.

f.

×	4	5	6
2	8	10	12
3	12	15	18

g.

×	3	4	5
7	21	28	35
8	24	32	40

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Answer Key _____ Date _____ Score _____ /10

Level 3 Operations Answer Key

I Can find even products

Number sense: A

1. Write the missing number. **10** more than **47** is **57** .
2. Write the missing number. **10** less than **20** is **10** .
3. Write <, > or = in the circle. **72** **>** **35**
4. Write how many of each place. **39** = **3** tens **9** ones
5. Write the missing number. **1** more than **33** is **34** .

Number sense: B

6. Write the missing number. **10** more than **877** is **887** .
7. Write the missing number. **10** less than **305** is **295** .
8. Write <, > or = in the circle. **544** **<** **565**
9. Write how many of each place. **84** = **8** tens **4** ones
10. Write the missing number. **1** more than **103** is **104** .

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I Can find even products

×	6	7	8	9
4	24	28	32	36
5	30	35	40	45

1 Shade the even rows.

The even row numbers: 4. Shade every box in that row.

×	6	7	8	9
4	24	28	32	36
5	30	35	40	45

2 Shade the even columns too.

The even column numbers: 6, 8. Shade those boxes too.

×	6	7	8	9
4	24	28	32	36
5	30	35	40	45

3 Both odd? Leave it white.

Both numbers odd? Leave the box white. The even products: 24, 28, 30, 32, 36, 40.

Say: "An even row or column: the product is even."

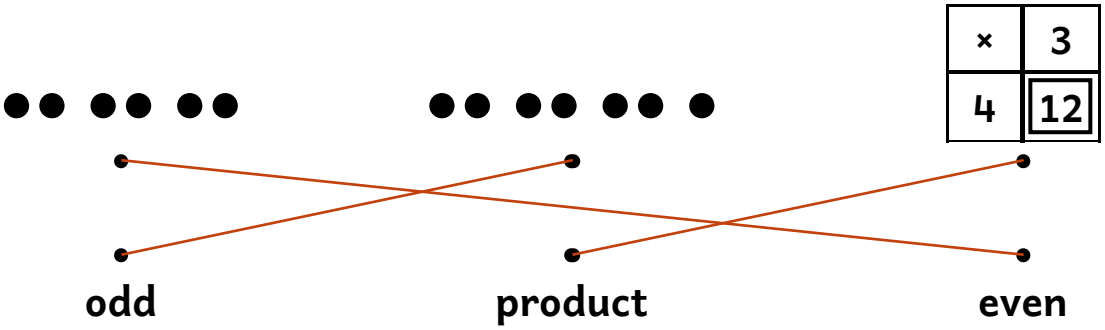
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Answer Key _____ Date _____ Score _____ /2

Level 3 Operations Answer Key

I Can find even products

Vocabulary: Draw a line to match.



Remember: An even row or column: the product is even.

Guided Practice: Shade every even product.

Steps: (1) Shade the even rows. (2) Shade the even columns too. (3) Both odd? Leave it white.

×	5	6	7	8
11	55	66	77	88
12	60	72	84	96

×	2	3	4	5
2	4	6	8	10
3	6	9	12	15

Independent Practice: Shade every even product.

a.

×	4	5	6	7
8	32	40	48	56
9	36	45	54	63

b.

×	6	7	8	9
2	12	14	16	18
3	18	21	24	27

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Answer Key _____ **Date** _____ **Score** _____ /6

Level 3 Operations Answer Key

I Can find even products

Steps: (1) Shade the even rows. (2) Shade the even columns too.
(3) Both odd? Leave it white.

Shade every even product.

a.

×	2	3
5	10	15
6	12	18
7	14	21

b.

×	2	3	4
7	14	21	28
8	16	24	32

c.

×	2	3	4
4	8	12	16
5	10	15	20

d.

×	4	5	6
7	28	35	42
8	32	40	48

e.

×	7	8	9
6	42	48	54
7	49	56	63

f.

×	11	12
4	44	48
5	55	60
6	66	72

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Answer Key _____ **Date** _____ **Score** _____ /1

Level 3 Operations Answer Key

I Can find even products

Read the story. Draw it. Solve.

There are 4 bags.

Each bag has 7 apples.

How many apples? Is it even or odd?

There are 28 apples.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

4 × 7 = 28

Say: "Is 4 or 7 even? The product is even ."

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Answer Key _____ Score _____ /3

Level 3 Operations Exit slip

I Can find even products

Shade every even product.

a.

×	5	6	7
5	25	30	35
6	30	36	42

b.

×	2	3	4
11	22	33	44
12	24	36	48

c.

×	7	8	9
8	56	64	72
9	63	72	81

☐ I can do this. ☐ I need one more day.

Answer Key _____ Score _____ /3

Level 3 Operations Exit slip

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☐ I can do this. ☐ I need one more day.

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I Can find even products

Level 3
Operations
Answer Key

Prerequisite routing:

Question	Prerequisite (T2 atom)	If missed → Lesson
a. Even numbers	Category, atom 1	2.OA.C.3 Even numbers: count by 2s
b–e. Facts of 2 and 4	Fact, atom 2	3.OA.C.7 Multiply by 2 and 4
f–g. Read the chart	Fact, atom 3	3.OA.C.7 Read the multiplication chart
	Rule	0 missed: go on · 1: that lesson · 2+: the earliest

Accepted answers:

- A shade drawn as a hatch, a fill or a ring around the number.
- Word problem: the product and "even"; the drawing may be groups or equal parts.
- A box shaded by mistake and crossed out: mark the pupil's last choice.
- A window is right only when every box is right: shaded even, white odd.
- The Say lines are spoken, not marked: listen for "row or column" and "even".

Answers at a glance:

Page	Answers	Fade
P0 Prerequisite	**14, 16 · 14 · 12 · 16 · 36 · 8, 12, 12 · 21, 35, 40**	★★
P1 Number sense	**57 · 10 · > · 3 tens 9 ones · 34 · 887 · 295 · < · 8 tens 4 ones · 104**	★★
P3 Guided	**60, 66, 72, 84, 88, 96 · 4, 6, 8, 10, 12**	★
P3 Independent	**32, 36, 40, 48, 54, 56 · 12, 14, 16, 18, 24**	★★
P4-1 Practice, rows 1–2	**10, 12, 14, 18 · 14, 16, 24, 28, 32 · 8, 10, 12, 16, 20 · 28, 32, 40, 42, 48**	★★
P4-1 Practice, last row	**42, 48, 54, 56 · 44, 48, 60, 66, 72**	★★★
P5 Word problem	**4 × 7 = 28**	★
P6 Exit slip	**a. 30, 36, 42 · b. 22, 24, 36, 44, 48 · c. 56, 64, 72**	★★★

Wrong answers to look for when marking:

Pupil wrote	Likely mistake	Next step
24, 36, 48	**Row only**	"Row OR column?" Finger on every even header.
56, 64, 72, 63, 81	**Odd × odd = even**	Count the pairs: one is left over.
22 left white (11 is odd)	**Any odd factor is odd**	One even number is enough.
30 left white (its tens digit is odd)	**Tens digit decides**	Cover the tens. Read the ones.
The even columns shaded, the 6 row white	**Columns only**	Rows first, then columns: finger on each header.
Word problem: 28, "odd"	**Any odd factor is odd**	Find one even number: 4. So 28 is even.
Word problem: 11	**Added the numbers**	4 groups of 7: draw the groups, then multiply.

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